DATE: 12/9/26

EXP-9 Implementation of Subnetting in cisco packet trace.

Aim:-

To Implementation of SUBNETTING IN CISCO PACKET TRACER simulator and test connectivity between different Subnets.

Requirements:

cisco packet Tracer

Routers

Switches

Pcs

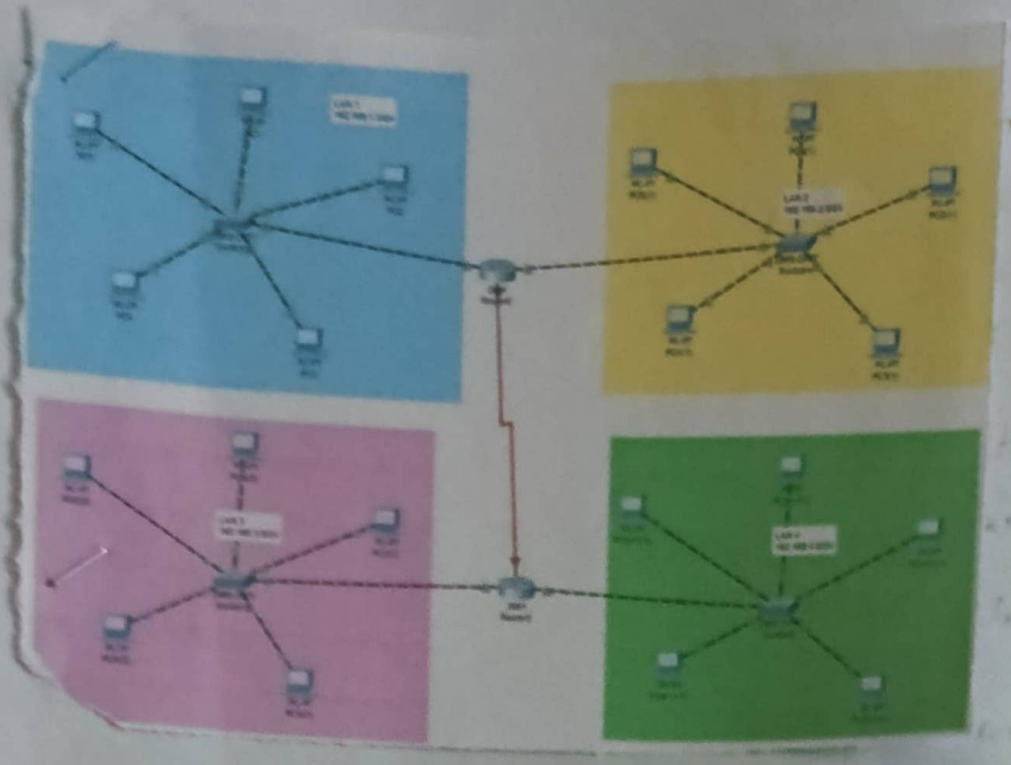
ethernet cables

Procedure:

1. creating the network Topology
   open cisco packet Tracer
   create a new workspace
   Add the required routers, switches & Pcs
   connect the devices using appropriate ethernet cables.

2. IP Addressing :-

LAN 1	R1G0/0	192.168.1.1/27
LAN 2	R1G0/1	192.168.2.1/27
LAN 3	R1S0/0	192.168.3.1/27
LAN 4	R2S0/0	192.168.4.1/27



For example,

IP Address : 192.168.1.11

Subnet mask : 225.255.255.224

Default gateway : 192.168.1.1

3. configuring Router P1

open R1 →
enables
configure terminal
interface gigabit ethernet 0110
IP addresses 192.168.1.1.225.255.244
NO shutdown.
exit.

4. configuring Router P2

enable
configure terminal
interface fast ethernet 0/10
IP address 192.168.31.255.255.224
no shutdown
exit.

5. configuring the ethernet

for S1:

enable
config terminal
interface fast ethernet 0/1
Switch port node access
exit

6. configuring the Pcs:-

open each Pc → Desktop → IP configuration and ethernet the appropriate IP address Subnet mask and default gateway.

Pc	IP address	Subnet mask	Default gateway
Pc_1	192.168.1.11	225.255.244	192.168.1.1
Pc_2	192.168.1.12	255.255.244	192.168.1.1